SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE:CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 50

Question Count: 4

Code of Conduct

I SIRANJEEVI G(192572166) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siranjeevi G

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

Permitted Tools :

draw.io / diagrams.net, Lucidchart, MindMeister, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool

Procedure for Preparing the Digital Concept Map

1. Use the following procedure to prepare the diagram and upload evidence.
2. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.
3. Open https://app.diagrams.net/ or the desktop version of draw.io / diagrams.net.
4. Choose Device, Google Drive, or OneDrive as the save location according to availability.
5. Create a blank diagram and name the file as CSA15_CO1_AT1_RegisterNumber_Concept_Map.drawio.
6. Place the concept map title at the center or as a clearly visible anchor node.
7. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
8. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
9. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for concept map title-specific elements.
10. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
11. Paste the exported diagram image or screenshot in this DOCX under Task 1.
12. Upload the DOCX, exported PNG/PDF, and .drawio file to Google Classroom.

Task 1: Concept Map

Create a concept map in draw.io, Lucidchart, Miro, Figma, or any suitable diagramming tool. The map must connect cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the student's Project.

Task 2A: Service Model Responsibility Matrix

Prepare a technical matrix that compares SaaS, PaaS, IaaS, FaaS/serverless, database-as-a-service, storage-as-a-service, identity-as-a-service, and monitoring/logging services. For each selected service category, explain what is managed by the cloud provider, what is managed by the student application team, and why that split fits the Project.

Task 2B: Cloud Service Decision Matrix

Map Project modules to cloud services. Include authentication, frontend hosting, backend/API layer, database, object storage, analytics/logging, notification, AI/Python module, security, CI/CD, backup, and monitoring. Students should justify the most relevant service choices instead of forcing every possible cloud service into the design.

Task 3: API Flow, Elasticity, and Scalability Design

Draw a professional cloud-flow diagram for the Project modules. The diagram must show user app or web app, API gateway/backend service, authentication, compute layer, database, object storage,

analytics/logging, monitoring, security boundaries, and scaling points. Students may replicate the given visual pattern, but all components and labels must be customized to their own architecture.

Task 4: Technical Justification

Write a concise justification explaining how the diagram and matrices prove CO1. The justification must discuss service model selection, API role, elasticity, scalability, virtualization or managed compute choice, and cloud responsibility boundaries.

1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Expected Digital Evidence

- Editable diagram source file and exported PNG/PDF for the concept map.
- Completed service model responsibility matrix and cloud service decision matrix.
- Architecture/API flow diagram exported as PNG/PDF.
- README file containing title, tools used, repository structure, and a short CO1 reflection.

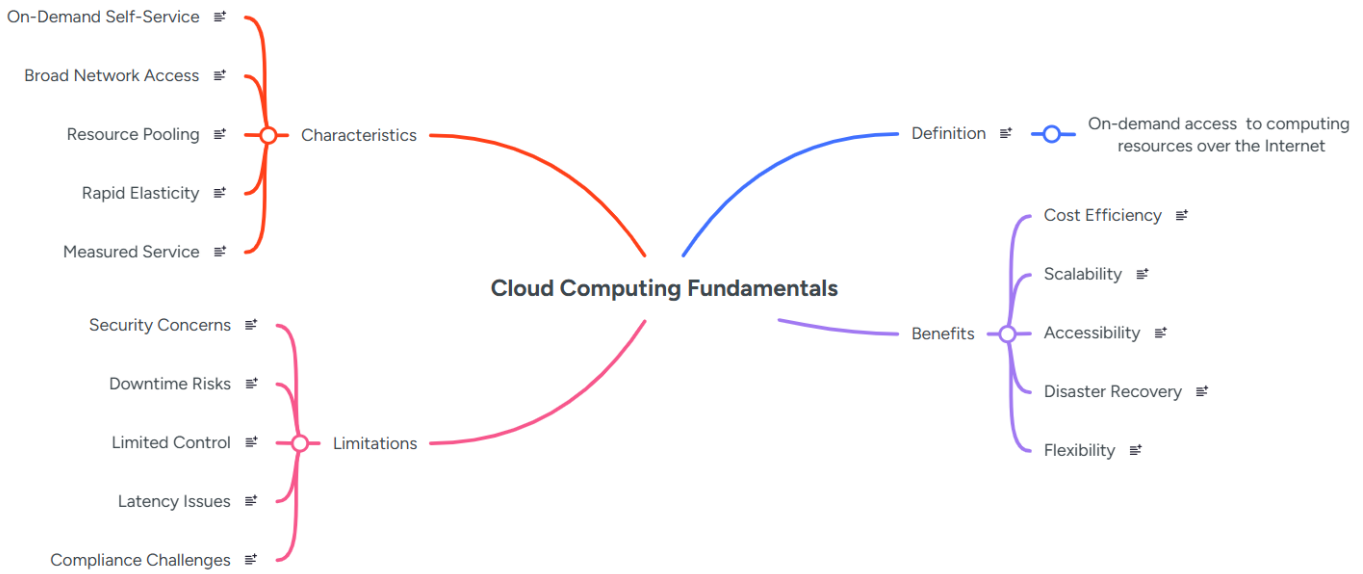
GitHub and Google Classroom Submission

1. Create a repository using a clear name: CSA15_CO1_AT<Number>_<RegisterNumber>.
2. Upload editable source files, exported PDF/PNG evidence, screenshots, and a short README explaining the work.
3. Keep filenames professional and include the register number where appropriate.
4. Submit only the GitHub repository link in Google Classroom before the deadline.
5. Ensure the repository is accessible to the faculty evaluator during assessment.

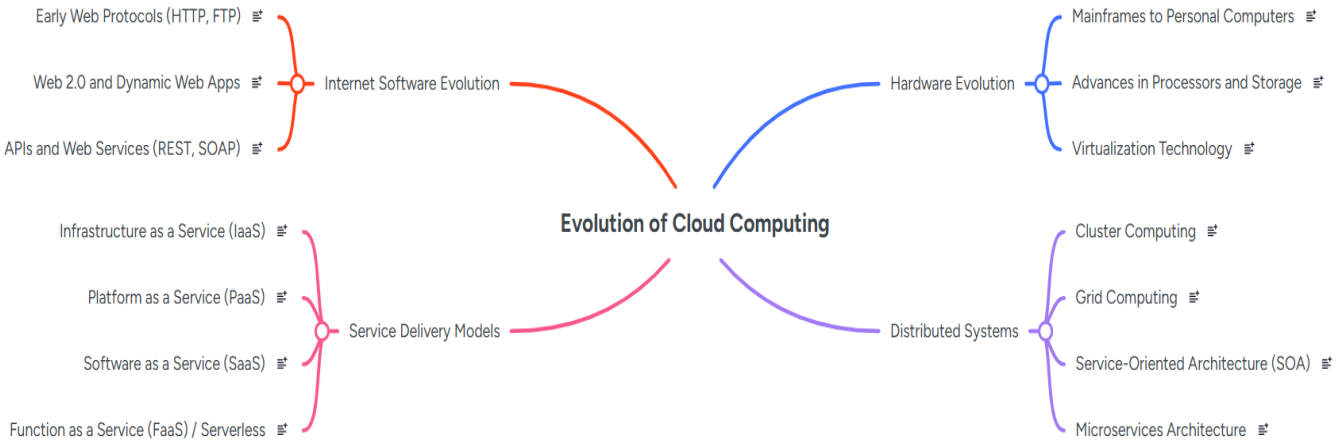
Implementation Note

Faculty may evaluate the submitted evidence digitally using repository inspection, diagram quality, oral explanation, and CO-aligned rubric scoring. The focus is on technical understanding, cloud design reasoning, and hands-on evidence rather than routine written work.

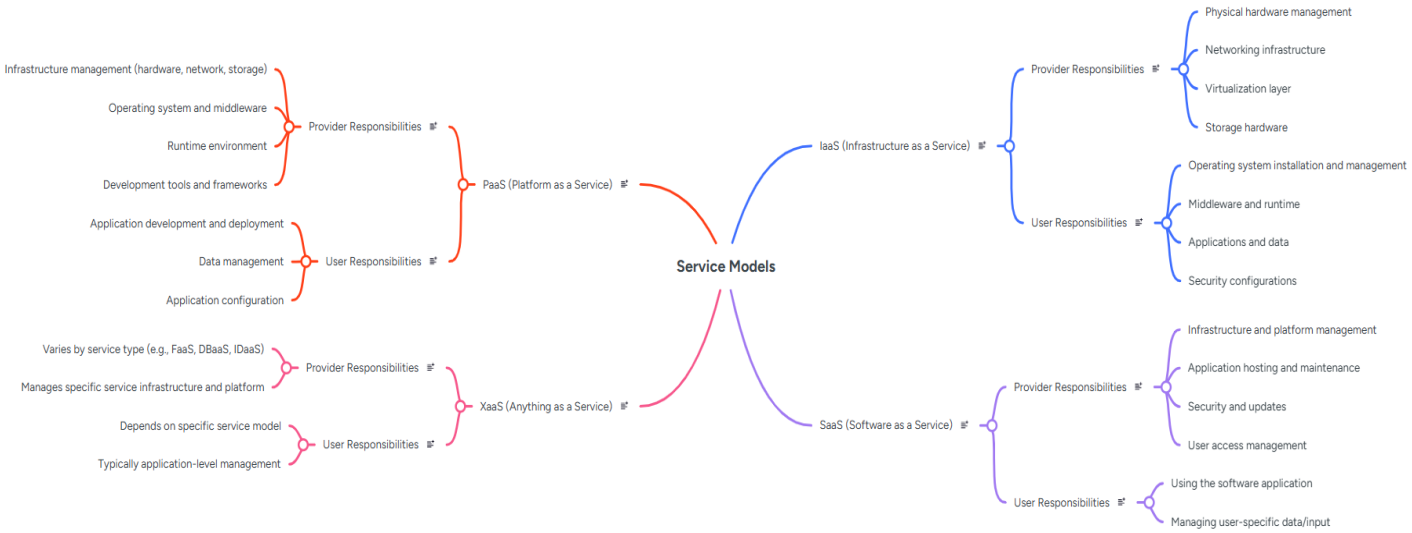
1. **Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.**



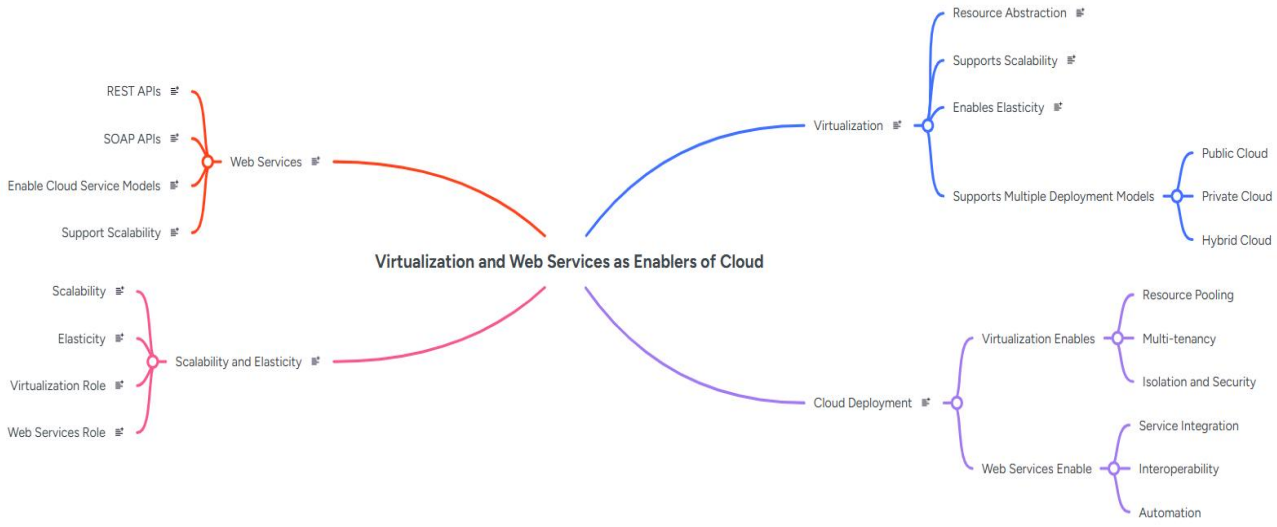
2. **Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.**



3. **Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.**



4. **Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability**



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

Cloud computing is the on-demand delivery of computing resources such as servers, storage, databases, networking and software over the Internet on a pay-as-you-go basis. In the concept map, Cloud Computing is the central node. A cloud provider offers these resources and a cloud consumer uses them, so the definition branch links the two actors through the service that connects them.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Cloud computing exhibits five key characteristics, as defined by NIST. On-demand self-service lets users provision resources without human interaction with the provider. Broad network access makes services available from any device over the network. Resource pooling, enabled by virtualization, serves many consumers from shared physical resources. Rapid elasticity adjusts resources to match demand, and scalability handles long-term growth. Measured service meters usage, which enables pay-per-use billing.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

These characteristics lead to several benefits. Pay-per-use billing gives cost savings, elasticity gives flexibility and agility, and scalability gives better performance. Resource pooling gives efficient resource utilization, broad network access gives global accessibility, and the provider's redundant infrastructure gives reliability and disaster recovery.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

Cloud computing also faces limitations. These include security and privacy risks, vendor lock-in, dependence on the Internet, limited control over the underlying infrastructure, compliance and data-location issues, and the possibility of downtime and latency.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

The branches are also connected by cross-links. Elasticity reduces over-provisioning cost, virtualization supports resource pooling, and the shared responsibility model mitigates security risks, while vendor lock-in reduces flexibility. Hierarchically, Level 1 is Cloud Computing, Level 2 is the four branches (Definition, Characteristics, Benefits and Limitations), and Level 3 is the leaf nodes. Colours are used meaningfully: blue for the title and branch nodes, green for characteristics, orange for benefits and red for limitations.